Copernicus Marine Service

# QUALITY INFORMATION DOCUMENT

# Baltic Sea Production Centre BALTICSEA_MULTIYEAR_BGC_003_012

**Issue:** 3.1

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**Approval date by the CMEMS product quality coordination team: 28/04/2025**

# CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

| Issue | Date | § | Description of Change | Author | Validated By |
|---|---|---|---|---|---|
| 1.0 | January 2018 | All | Creation of the document | Lars Axell | |
| 1.1 | 5.4.2019 | All | New data and include 2017 data in validation period | Svetlana Verjovkina | |
| 2.0 | 12.4.2019 | All | New data:<br>- 2012-2016 rerun (due to bug)<br>2017 data included | Simon Jandt | Mercator Ocean |
| 2.1 | 3.7.2019 | All | Minor editing and addition of update plans | Fabian Schwichtenberg, Ina Lorkowski | |
| 2.2 | 16.08.2019 | All | Minor editing | Simon Jandt-Scheelke | |
| 2.3 | 23.08.2019 | All | Replacing figures for innovation statistics<br>Minor editing | Lars Axell, Ye Liu | |
| 2.4 | 02.09.2019 | All | Proof reading | Vibeke Huess | |
| 2.5 | 10.09.2019 | All | Editing | Ye Liu<br>Simon Jandt-Scheelke | Mercator Ocean |
| 3.0 | 14.11.2021 | All | Update for new MY product V202303 | Lena Spruch | |
| 3.0 | 19.01.2023 | All | Update for new MY product V202303 | Lena Spruch, Svetlana Verjovkina | V. Huess |
| 3.1 | Feb 2025 | | Update of product extent information | V. Huess | V.Huess |

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# I EXECUTIVE SUMMARY

## I.1 Products covered by this document

The product BALTICSEA_MULTIYEAR_BGC_003_012 (Product reference: https://doi.org/10.48670/moi-00012), is a Baltic Sea biogeochemical reanalysis covering the period January 1993 and up to as close as possible to present. The product provides daily, monthly and yearly mean fields for the following variables:

- dissolved oxygen
- ammonium
- nitrate
- phosphate
- chlorophyll-a
- Secchi depth (noon values)
- net primary production (daily accumulated values)
- pH
- partial pressure of $CO_2$

The BAL MFC model system that was used to produce the reanalysis consists of the biogeochemical model ERGOM, which is online-coupled to the physical model NEMO. The model system produces 3D time series, covering the area of the Baltic Sea with a horizontal resolution of 1 nautical mile. Vertically, it provides 56 depth layers varying between 1 – 24 m.

This reanalysis was released into the Copernicus Marine Service Product Catalogue in March 2023.

The product in the Catalogue is extended in time on a yearly basis providing a homogenous time series of data. This "final dataset" will be available up to approximately present time minus 1-2 years; depending on the release of the needed forcing data. To provide a more "close to present" dataset we will, from Spring 2025 and onwards, additionally provide a monthly extension of the product with a coverage up to present time minus two months. These extensions, called an interim product, will be run by the same model system, but with the available forcing dataset at production time. These interim products will therefore not provide an exact homogenous dataset over the full product time period. The interim product extensions will be replaced in the Product Catalogue by a final product extension when the forcing data are ready (approximately 1-2 years later). Whether a product file is from an interim or final production is indicated in the file name.

For the quality assessment of the BALTICSEA_MULTIYEAR_BGC_003_012 product, the reanalysis period from 1993 to 2021 is validated with observational data from monitoring stations, ferryboxes and satellite products.

## I.2 Summary of the results

**Mixed layer depth (MLD):** Calculation of the mixed layer depth is based on the gradient of the water density. From observations, the mixed layer depth is derived from salinity and temperature profiles of ICES-HELCOM observations. These data were compared with the new reanalysis product. During whole validation period the product shows shallower MLD than observations.

**Dissolved oxygen**: At the surface, dissolved oxygen concentrations and seasonal dynamics are reproduced well by the model across all regions of the Baltic Sea. In deeper layers, there is a stronger positive bias, especially in deep basins where conditions are typically anoxic.

**Ammonium:** There is a reasonable agreement between the modelled ammonium concentrations and the observational data at the surface, with a slight negative bias. With depth, the negative bias increases and in very deep, anoxic layers ammonium concentrations can be considerably underestimated by the model.

**Nitrate:** The seasonal dynamics of nitrate concentrations at the surface are well reproduced by the model simulation. Near the surface, there is a slight negative bias for nitrate concentrations. In deeper layers, especially in anoxic regions, the simulation often overestimates nitrate concentrations.

**Phosphate:** There is a reasonable agreement between the model simulation and observations, with a slight positive bias at the surface. The seasonal dynamics of phosphate concentrations at the surface are reproduced well by the simulation. In the deeper layers of the water column, phosphate concentrations are underestimated by the model, in particular in regions that are anoxic.

**Chlorophyll a:** The chlorophyll a concentrations simulated with the model are lower than the observations in most regions of the Baltic Sea. The seasonal dynamics are simulated well, but the magnitude of the spring bloom is underestimated in some years.

**Secchi depth:** The simulated values for Secchi depth are in reasonable agreement with the measurements. The typical seasonal patterns of Secchi depth are reproduced, however Secchi depth is overall overestimated by the model simulation.

**Surface pCO2**: The partial pressure of carbon dioxide (pCO2) at the surface of the reanalysis product is validated with measurements from the in-situ ferrybox observations obtained from the SOCAT database. The product annually overestimates the partial pressure of carbon dioxide in spring and early summer, during winter and autumn the product annually shows slightly smaller pCO2 than observations.

**pH:** The validation results indicate that the model simulates the general seasonal and spatial patterns of pH, whereas the pH values at the surface are matched better by the model than in deeper layers of the water column.

## I.3 Estimated Accuracy Numbers

Table I.1 summarises the Estimated Accuracy Numbers (EAN), which give an overview of the quality of the model results, for each variable of the biogeochemical reanalysis product for the Baltic Sea. The EANs are the root mean square deviation (RMSD) of the results from the 28-year reanalysis period in relation to in-situ measurements at monitoring stations. The RMSD values for different depth levels shown in Table I.1 represent averages over the validation period from 01.01.1993-31.12.2018 and all monitoring stations. It has to be noted, that the in-situ measurements used for the calculation of the EANs are not evenly distributed in time and only cover a very small fraction of the model domain. Therefore, they represent a "snapshot" of the model quality, and might not be representative for the whole model domain.

*Table I.1 EANs for the biogeochemical reanalysis product for the Baltic Sea. Values represent averages of all monitoring stations for 5 depth layers and the validation period from 1.1.1993 to 31.12.2018.*

| RMS deviation (RMSD) | water column | 0-5 m depth | 5-30 m depth | 30-80 m depth | 80-200 m depth | 200-400 m depth |
|---|---|---|---|---|---|---|
| Chlorophyll-a [mg $m^{-3}$] | - | 2.98 | 2.13 | 1.98 | - | - |
| Oxygen [mmol $m^{-3}$] | - | 27.28 | 29.78 | 63.78 | 92.59 | 65.05 |
| Nitrate [mmol $m^{-3}$] | - | 1.88 | 1.86 | 3.22 | 7.60 | 15.33 |
| Phosphate [mmol $m^{-3}$] | - | 0.19 | 0.22 | 0.53 | 1.21 | 2.26 |
| Ammonium [mmol $m^{-3}$] | - | 0.29 | 0.70 | 0.96 | 3.87 | 20.34 |
| Secchi Depth [m] | 3.17 | - | - | - | - | - |
| pH | - | 0.29 | 0.25 | 0.33 | 0.49 | 0.57 |

# II PRODUCTION SYSTEM DESCRIPTION

**Production centre name:** Danish Meteorological Institute (DMI)

**Production system name:** Baltic Sea reanalysis biogeochemistry (BALTICSEA_MULTIYEAR_BGC_003_012)

**Lateral product boundaries:** 53°N – 65.9°N, 9.04°E – 30.21°E
**Spatial resolution**:

- Latitude: 0.01666°
- Longitude: 0.027777°
- Depth layer (depth of the midpoint of each bin): 0.50, 1.52, 2.55, 3.60, 4.68, 5.80, 6.96, 8.17, 9.45, 10.81, 12.27, 13.86, 15.60, 17.53, 19.70, 22.15, 24.94, 28.15, 31.86, 36.15, 41.13, 46.91, 53.59, 61.28, 70.08, 80.07, 91.31, 103.82, 117.62, 132.67, 148.90, 166.25, 184.60, 203.86, 223.92, 244.66, 265.99, 287.81, 310.05, 332.64, 355.51, 378.61, 401.91, 425.36, 448.93, 472.61, 496.36, 520.19, 544.06, 567.98, 591.94, 615.92, 639.92, 663.94, 687.98, 712.02)

**Maximum number of model depth levels:** 56

## II.1 Model system

The BAL MFC model system consist of four building blocks:

(1) The **3D ocean-ice model Nemo** (Nucleus for European Modelling of the Ocean), which is an open community ocean model (https://nemo-ocean.eu). In the BAL MFC model system the Nemo version 4.0 is used in combination with the sea ice and thermodynamic model SI3. Our Nemo configuration and set-up is adapted for the Baltic Sea.
(2) The **biogeochemical model ERGOM** (Ecological Regional Ocean Model, https://ergom.net/), which is developed at IOW (Germany) to model biogeochemical cycles in the Baltic Sea. The version used in the BAL MFC model system is based on Neumann et al., 2015*: A new radiation model for Baltic Sea ecosystem modelling. Journal of Marine Systems 152: 83-91.* The model describes the basic nutrient and carbon cycles through 15 pelagic state variables. The sediment is not vertically resolved and consists of two organic nutrient components.
(3) The **wave model WAM**, an open source model developed at Hereon (Germany). The WAM version used in the BAL MFC model system is based on WAM version 4.6.2., and has been adapted to the Baltic Sea. The model uses 36 spectral directions and 35 different frequencies between 0.04177 and 1.06719 Hz.
(4) The **data assimilation system PDAF** (Parallel Data Assimilation Framework https://pdaf.awi.de/), used with the LESTKF filter.

The Nemo and ERGOM models are run one-way online-coupled and use the PDAF system to assimilate observations into the analysis field in Nemo. The WAM model runs independently from the Nemo-ERGOM-PDAF system. Figure II.1 shows an overview of the complete BAL MFC model system set-up and *Figure* II.2 shows a detailed overview of ERGOM.

Both model systems run with a horizontal resolution of 1 nautical mile and use the same bathymetry grid based on GEBCO v2013 data. Both model systems are forced with the same best available atmospheric forcing dataset.

The native model domain for the Nemo-ERGOM system covers the North Sea and Baltic Sea with open boundaries at Scotland-Norway and the English Channel. The wave system's native model domain covers the Baltic Sea and towards the eastern part of the North Sea (from 8 degree E).

Our production with the Nemo v.4.0 system is using the TEOS-10 equation of state for the model calculations. The temperature and salinity input data are converted to absolute salinity and conservative temperature, and the final products are calculated back to practical salinity and potential temperature.

## II.2 Multi year (MY) production system

The BAL MFC multi year products (BALTICSEA_MULTIYEAR_PHY_003_011 and BALTICSEA_MULTIYEAR_BGC_003_012) are produced with the online one-way coupled Nemo-ERGOM model. The BAL MFC multi year product for the wave conditions (BALTICSEA_REANALYSIS_WAV_003_015) is produced with the WAM system. These two multi year production systems are for the time being run without any interaction (i.e. with no exchange of parameters). The ERA5 dataset available from Copernicus Climate Service is used as atmospheric forcing data for both production systems. The products cover the period from 1/1 1993 up till as close to present as possible (approximately present time – 1 Y).

**Details for the Nemo-ERGOM production:**

The following parameters are used as atmospheric forcing: 10 m wind speed and direction, surface temperature, air pressure at sea surface, solar radiation total precipitation, air humidity. For the atmospheric deposition of nitrate and ammonium, ERGOM uses the yearly data set from EMEP (https://emep.int/mscw/mscw_moddata.html). The atmospheric concentration of $CO_2$ is estimated from the NOAA global monthly mean $CO_2$ data (https://gml.noaa.gov/ccgg/).

Information about river discharges and the rivers water temperature are included in the Nemo-ERGOM system from the historical data set of the EHYPE-model system (version: ehype3_fgd produced at SMHI, Sweden in 2020). This data set also contains nutrient loads for nitrogen (inorganic and total) and phosphorus (soluble reactive and total) used by ERGOM. River loads of silicate, dissolved inorganic carbon and alkalinity are based on Pätsch and Lenhart (2004) and the Baltic database. Data from 849 rivers are included (for the native North Sea – Baltic Sea full domain).

At the lateral boundaries, daily mean values from the CMEMS North West Shelf multi-year product are used (NWSHELF_MULTIYEAR_PHY_004_009, 10.48670/moi-00059). From this model, the following parameters are used: Sea level, temperature, salinity, horizontal velocities. Additionally, hourly tidal information is added at the open boundaries based on 9 tidal constituents from a tidal model.

For the biogeochemical model ERGOM interpolated data from ICES for DIN, DIP, silicate and oxygen are used at the lateral boundaries (https://www.ices.dk). Zoo- and phytoplankton data are based on ICES observations of chlorophyll-a (chlorophyll to plankton biomass). The parameters of the carbonate system Total Alkalinity (TA) and Dissolved Inorganic Carbon (DIC) are based on Wolf-Gladrow et al. (2007, https://doi.org/10.1016/j.marchem.2007.01.006) and results of the biogeochemical model ECOHAM (Lorkowski et al., 2012, https://doi.org/10.1016/j.ecss.2011.11.037).

Satellite observations of SST and in-situ profile observations are assimilated into the Nemo analysis field using the PDAF tool at every midnight. The assimilation is done in a three day cycle, alternating between the assimilation of SST, temperature profiles and salinity profiles. All variables are assimilated separately in a uni-variate sense. Increments are added to the model forecast state us-

ing Incremental Analysis Update (IAU) module with the time window of 24 hours. The satellite observation are based on the Copernicus Marine Service L3S SST multi year observations. The in-situ profile observations are from the ICES database. A climatologic ensemble with 75 members is used for the assimilation. The ensemble is based on a 5 year (2011-2015) free run of the Nemo system and created with a 15-day moving window.

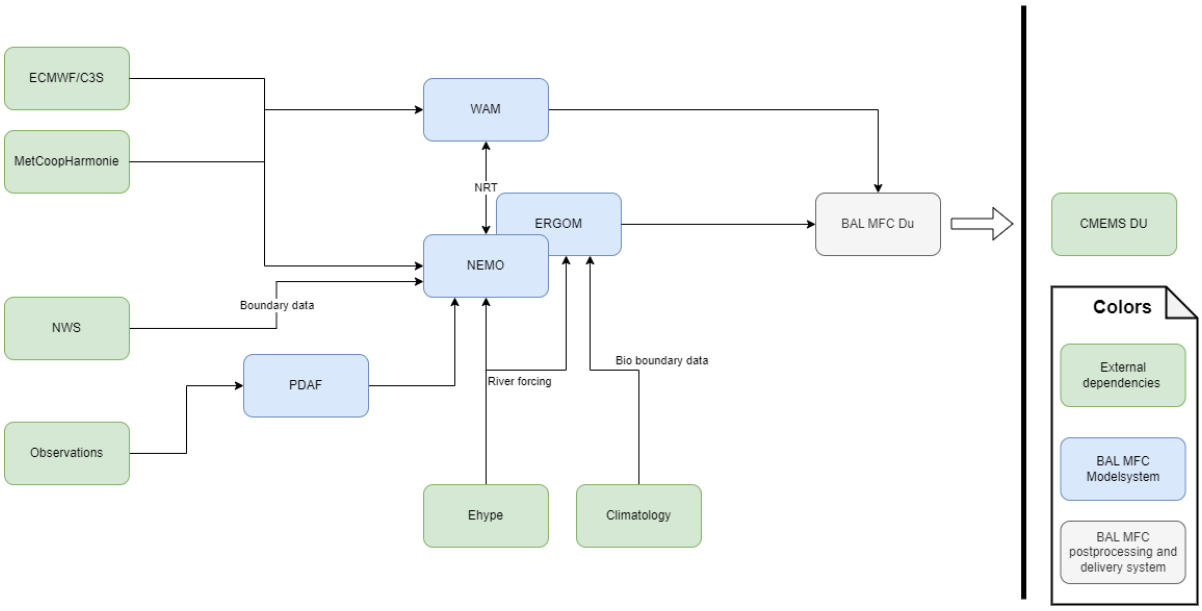


*Figure II.1 System overview of the BAL MFC model system.*

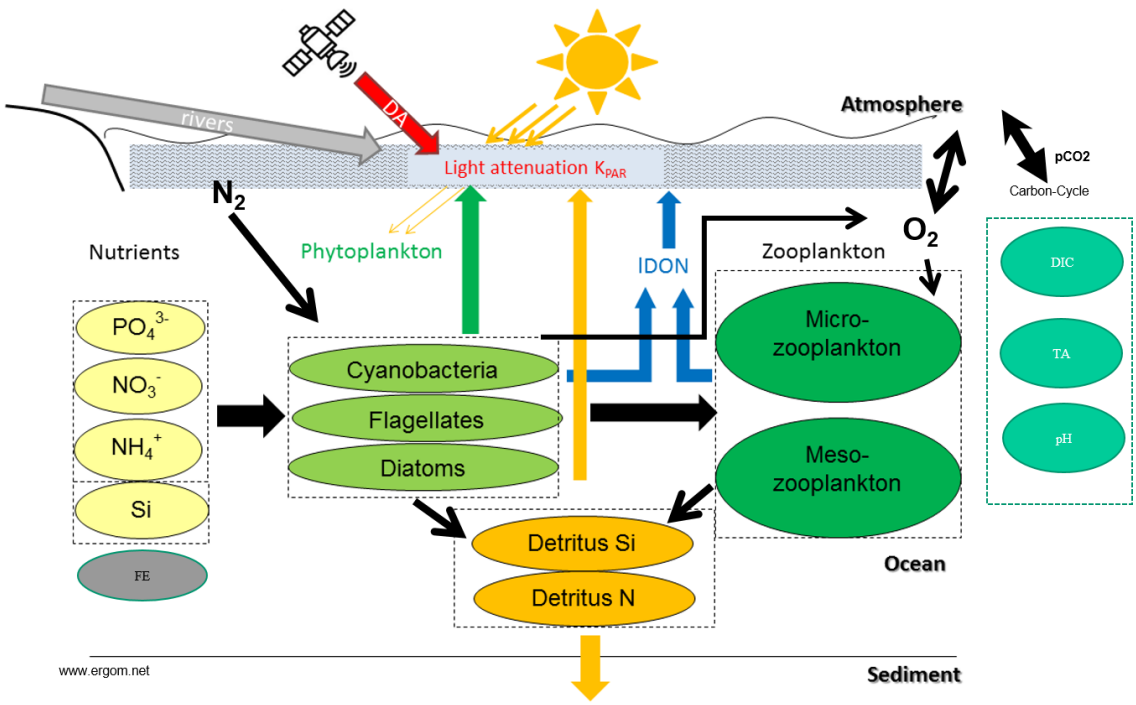


*Figure II.2 schematic overview of the ERGOM, the biogeochemical component of the BAL MFC model system.*

# III VALIDATION FRAMEWORK

To assess the quality of the BALTICSEA_MULTIYEAR_BGC_003_012 products, the simulation results of the reanalysis run from 01.01.1993 to 31.12.2021, are compared to observational data. This includes in-situ measurements, satellite observations and ferrybox data. The availability of observations varies strongly for the simulated biogeochemical variables (see Table III.2). For net primary production, no observational data is available for the simulated period, therefore it is not validated. In addition to the biogeochemical variables, we also validate mixed layer depth, as temperature and salinity distributions have a significant effect on biogeochemical processes and the distribution of biogeochemical variables in the water column.

*Table III.2 Overview of available observations for each biogeochemical variable*

| Variable | Ferry box data | In-situ measurements | Satellite data |
|---|---|---|---|
| **Dissolved oxygen** | - | HELCOM stations (ICES database): profile data 1993-2021 | - |
| **Nutrients (nitrate, ammonium, phosphate)** | - | HELCOM stations (ICES database): profile data 1993-2021 | - |
| **Chlorophyll a** | - | HELCOM stations (ICES database): profile data 1993-2021 | OCEANCOL-OUR_BAL_BGC_L3_MY_009_133 1997-2021 |
| **pH** | - | HELCOM stations (ICES database): profile data 1993-2021 | - |
| **pCO2** | SOCAT database 2012 - 2021 | - | - |
| **Secchi depth** | - | HELCOM stations (ICES database): profile data 1993-2021 | - |
| **Net primary production** | - | - | - |

## III.1 Observations

The observational data that is available for the validation of the model results includes CTD data from monitoring stations and moorings, downloaded from the ICES database, measurements from ferryboxes and L3 satellite observations. The detailed validation procedures for each type of available observational data are described in the following.

**Monitoring stations and moorings**

Figure III.3 shows the geographical locations of all moorings and monitoring stations that are included in the assessment, as well as the geographical regions of the Baltic Sea. Moorings provide more continuous time series than monitoring stations. At each monitoring station, validation statistics for two depth layers are calculated: a surface layer and a deep layer. The evaluated layers represent the shallowest and deepest layer, in which more than four data points are available for the validation period.

Please be aware, that the shallowest and the deepest data points that are considered for each station are not necessarily at the surface or the bottom of the water column.

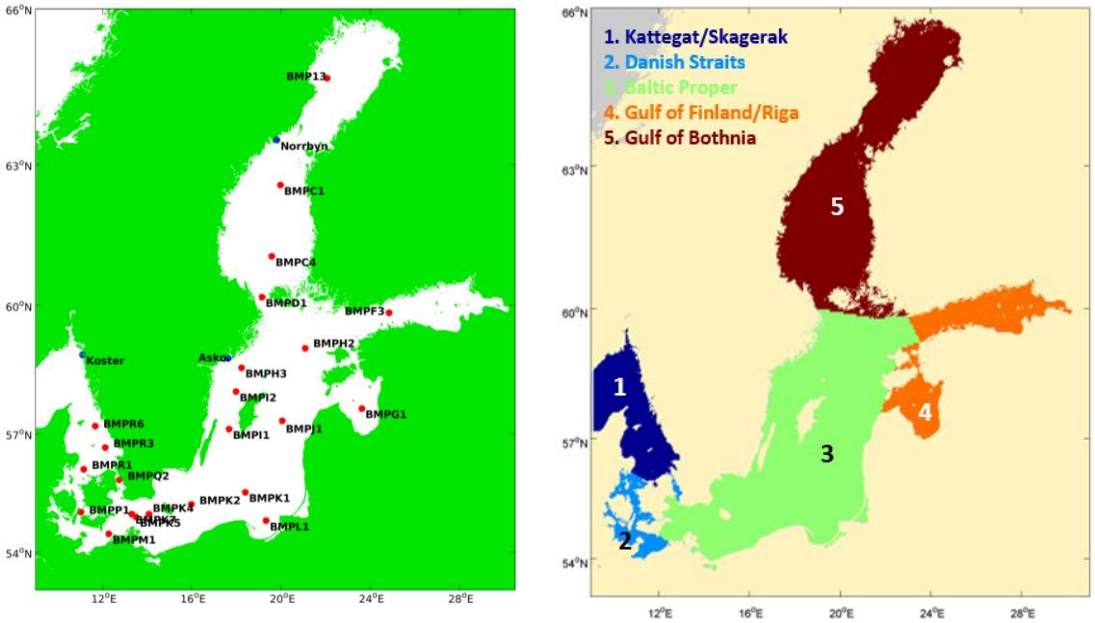


*Figure III.3 LEFT: All moorings (blue dots) and monitoring stations (red dots) that are accounted for in this assessment (left) all five sub regions, which are used for location of the monitoring stations and moorings (right). RIGHT: All five subregions of the Baltic Sea.*

**Ferrybox transects**

Ferryboxes provide near surface measurements with a high temporal resolution. The quality of ferrybox measurements varies and is dependent on biofouling at the water intake of the ferry box system. Ferry boxes deliver data along the track of ships, which provides region-wide information (see *Figure* III.4). However, since the routes of the ships can change from trip to trip, the available measurements at one specific location are rather scarcely distributed.

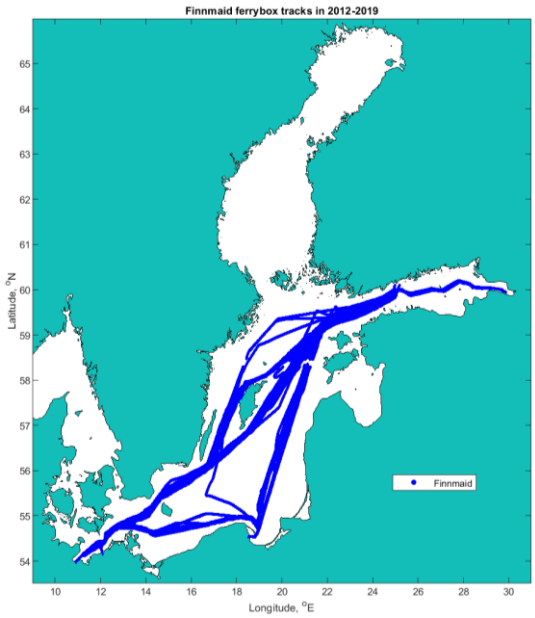


*Figure III.4 Locations of Ferry box pCO2 observations along the tracks of the ships for the period January 2012 to December 2019.*

**Satellite observations**

For chlorophyll a, L3 super collated satellite observations are available. The spatial coverage and quality of the satellite observations varies from time-step to time-step. However, these observations provide a good overview of the surface distribution of chlorophyll and the spatial and temporal dynamics of phytoplankton blooms.

## III.2 Statistical metrics

Suppose we have the discrete reference data $r(i,j,k,t)$ defined in time and space and we need to verify the product data $p(i,j,k,t,\mathrm{fcl},\mathrm{pid})$ against the reference data. The $i,j,k,t$ denote space and time indices, $\mathrm{fcl}$ the forecast length of the product and $\mathrm{pid}$ the product identification index. Therefore, we could have several products with several forecast lengths all verified against single reference data. *Table* III.3 defines the statistics as they are defined throughout the validation routines. The variable x denotes either *r* or *p*.

*Table III.3 Definition of the statistical metrics used for validation. The [stat] is the name of the statistics (i.e. mean, var, cov, MSE, corr), $N_t$ is the total number of time steps covered, the $N_i$, $N_j$, $N_k$ are the total number of spatial coverage and the $x$ is the variable, i.e. reference or product data.*

| **Variable name** | **Defining equation** | **Equation number** |
|---|---|---|
| Temporal [stat] | $[\text{stat}]_t \equiv \frac{1}{N_t}\sum_{t=1}^{N_t} x$ | (1) |
| Spatial [stat] | $[\text{stat}]_s \equiv \frac{1}{N_i}\frac{1}{N_j}\frac{1}{N_k}\sum_{i=1}^{N_i}\sum_{j=1}^{N_j}\sum_{k=1}^{N_k} x$ | (2) |
| Mean | $\overline{x} = \text{mean}(x) \equiv \text{mean}_s\big(\text{mean}_t(x)\big) = \frac{1}{N_i}\frac{1}{N_j}\frac{1}{N_k}\frac{1}{N_t}\sum_{i=1}^{N_i}\sum_{j=1}^{N_j}\sum_{k=1}^{N_k}\sum_{t=1}^{N_t} x$ | (3) |
| Variance | $\text{var}(x) \equiv \text{var_}s\big(\text{var_}t(x)\big) == \frac{1}{N_i}\frac{1}{N_j}\frac{1}{N_k}\frac{1}{N_t}\sum_{i=1}^{N_i}\sum_{j=1}^{N_j}\sum_{k=1}^{N_k}\sum_{t=1}^{N_t}\big(x - \text{mean}(x)\big)^2$ | (4) |
| Standard deviation | $\sigma_x = \text{std}(x) = \sqrt{\text{var}(x)}$ | (5) |
| Covariance | $\text{cov}(r,p) \equiv \text{cov_}s\big(\text{cov_}t(r,p)\big) == \frac{1}{N_i}\frac{1}{N_j}\frac{1}{N_k}\frac{1}{N_t}\sum_{i=1}^{N_i}\sum_{j=1}^{N_j}\sum_{k=1}^{N_k}\sum_{t=1}^{N_t}\big(r - \text{mean}(r)\big)\cdot\big(p - \text{mean}(p)\big)$ | (6) |
| Mean square difference | $\text{MSD}(p,r) \equiv \text{MSD}_s\big(\text{MSD}_t(p,r)\big) == \frac{1}{N_i}\frac{1}{N_j}\frac{1}{N_k}\frac{1}{N_t}\sum_{i=1}^{N_i}\sum_{j=1}^{N_j}\sum_{k=1}^{N_k}\sum_{t=1}^{N_t}(p-r)^2$ | (7) |
| Correlation | $R = \text{corr}(p,r) = \frac{\text{cov}(p,r)}{\sqrt{\text{var}(p)\cdot\text{var}(r)}} = \frac{\text{cov}(p,r)}{\sigma_p\cdot\sigma_r}$ | (8) |
| Root mean square difference | $\text{RMSD}(p,r) = \sqrt{\text{MSD}(p,r)}$ | (9) |
| Bias | $B = \text{bias}(p,r) = \text{mean}(p-r) = \text{mean}(p) - \text{mean}(r)$<br>$\text{SST: bias}_t(p,r) = \text{mean}_t(p) - \text{mean}_t(r)$ | (10) |
| Centred pattern root mean square difference | $\text{cRMSD}(p,r) = \text{RMSD}\big(p - \text{mean}(p), r - \text{mean}(r)\big) == \sqrt{\frac{1}{N_i}\frac{1}{N_j}\frac{1}{N_k}\frac{1}{N_t}\sum_{i=1}^{N_i}\sum_{j=1}^{N_j}\sum_{k=1}^{N_k}\sum_{t=1}^{N_t}\Big(\big(p - \text{mean}(p)\big) - \big(r - \text{mean}(r)\big)\Big)^2}$ | (11) |

The difference between cRMSD and RMSD

It can easily be shown that the square of RMSD is the sum of the squares of the cRMSD and Bias. Following the equations (9), (10) and (11) we get:

$$\mathrm{RMSD}^2 = \mathrm{MSD} = \frac{1}{N}\sum_{i=1}^{N}(p-r)^2$$

$$\mathrm{bias} = B = \overline{p} - \overline{r} = \frac{1}{N}\sum_{i=1}^{N} p - \frac{1}{N}\sum_{i=1}^{N} r = \frac{1}{N}\sum_{i=1}^{N}(p-r)$$

$$\mathrm{cRMSD}^2 = \mathrm{RMSD}^2(p-\overline{p}, r-\overline{r}) = \frac{1}{N}\sum_{i=1}^{N}\left((p-\overline{p})-(r-\overline{r})\right)^2 = \frac{1}{N}\sum_{i=1}^{N}\left((p-r)-(\overline{p}-\overline{r})\right)^2 ==$$

$$\frac{1}{N}\sum_{i=1}^{N}(p-r)^2 - 2\cdot(\overline{p}-\overline{r})\cdot\frac{1}{N}\sum_{i=1}^{N}(p-r) + \frac{N}{N}(\overline{p}-\overline{r})^2 = \mathrm{RMSD}^2 - 2\cdot B^2 + B^2 == \mathrm{RMSD}^2 - B^2 \tag{12}$$

$$\Rightarrow \mathrm{RMSD}^2 = \mathrm{cRMSD}^2 + B^2$$

The basis of Target diagram

The Target diagram is based on the relation between RMSD and the centralized RMSD (which is called "Standard deviation of bias" as well). (See equation 12.) (Jolliff et al., 2009)

$$RMSD^2 = cRMSD^2 + B^2$$

The three presented metrics are normalized by the standard deviation of the observations ${\sigma_r}^{-2}$.

$$\left(\frac{RMSD}{\sigma_r}\right)^2 = \left(\frac{cRMSD}{\sigma_r}\right)^2 + \left(\frac{B}{\sigma_r}\right)^2$$

Furthermore, the cRMSD is expanded by the information of the sign of the difference between the standard deviation of the product data and the standard deviation of the reference.

$$'(\sigma_D) = \quad '(\sigma_p - \sigma_r)$$

The following equation is represented in the target diagram:

$$\frac{RMSD^2}{\sigma_r} = \frac{(\quad '(\sigma_D)cRMSD)^2}{\sigma_r} + \frac{B^2}{\sigma_r}$$

The "Target Diagram" therefore shows the normalized bias, the normalized cRMSD and the sign of the difference of standard deviation between the model and observations.

# IV VALIDATION RESULTS

The following sections give an overview of the statistical validation results for each biogeochemical variable. In addition to the biogeochemical variables, validation results for physical variables, which have significant impact on the biogeochemical processes, are presented. Here, we only show a selection of the validation results, the complete set of validation plots for all variables is available on the BOOS website:

http://www.boos.org/cmems-baltic-mfc-product-quality-information-multi-year-product/

For the validation with in-situ data from monitoring stations, measurements from 10 stations were available for each variable. The validated stations cover three subregions of the Baltic Sea: the Baltic Proper, the Gulf of Finland and Riga, the Gulf of Bothnia. For an overview of the subregions and the geographical location of the monitoring stations, see Figure III.3.

## IV.1 Mixed layer depth (MLD)

Calculation of the mixed layer depth is based on the gradient of the water density. From observations, the mixed layer depth is derived from salinity and temperature profiles of ICES-HELCOM observations. These data were compared with the new reanalysis product. A list of ICES-HELCOM stations is found in Table IV.4. Along with the plots of time series of the MLD data for each station that are displayed in Figure IV.5 and Figure IV.6. Figure IV.7 shows the Target diagram.

Temperature and salinity profiles of ICES-HELCOM observations are rather scarce. The annual cycle of the MLD of the products follows the MLD calculated from observations for all reanalysis period. It is possible to observe the seasonal variability of the MLD product in summer and winter. During the summer, the MLD is lower, and during the winter, the MLD is deeper. In the Baltic Proper the mixed layer depth is small from late spring until end of summer reaching values of 10-20 m in period 01.01.1993-30.12.2019. In autumn and winter, MLD increases considerably with maximum MLD of 70 m. In the Gulf of Bothnia, the MLD varies from 10 to 80 m deep during reanalysis period. The MLD is small from late spring until end of summer reaching values of 10-15 m, and in autumn and winter, mixed layer depth increases considerably with maximum MLD of 80 m.

The cRMSD of MLD for V202303 reanalysis product is 11.34 m and correlation coefficient is 0.69 for the whole reanalysis period 1993-2019 in the bulk comparison of MLD product with observations.

*Table IV.4: Station list of ICES HELCOM profile station locations*

| QuIDV3name | ICES_NAME | Longitude °E | Latitude °N |
|---|---|---|---|
| BMP13 | F9 | 22,0628 | 64,7003 |
| BMPC1 | US5B | 19,9688 | 62,5862 |
| BMPG1 | 121 | 23,6167 | 57,6167 |
| BMPH2 | OMBMPH2 | 21,0833 | 59,0333 |
| BMPI2 | BY32 | 17,9833 | 58,0167 |
| BMPJ1 | BY15 | 20,0500 | 57,3200 |
| BMPK2 | BMPK2 | 15,9833 | 55,2500 |
| BMPK4 | OMBMPK5 | 13,5000 | 54,9250 |

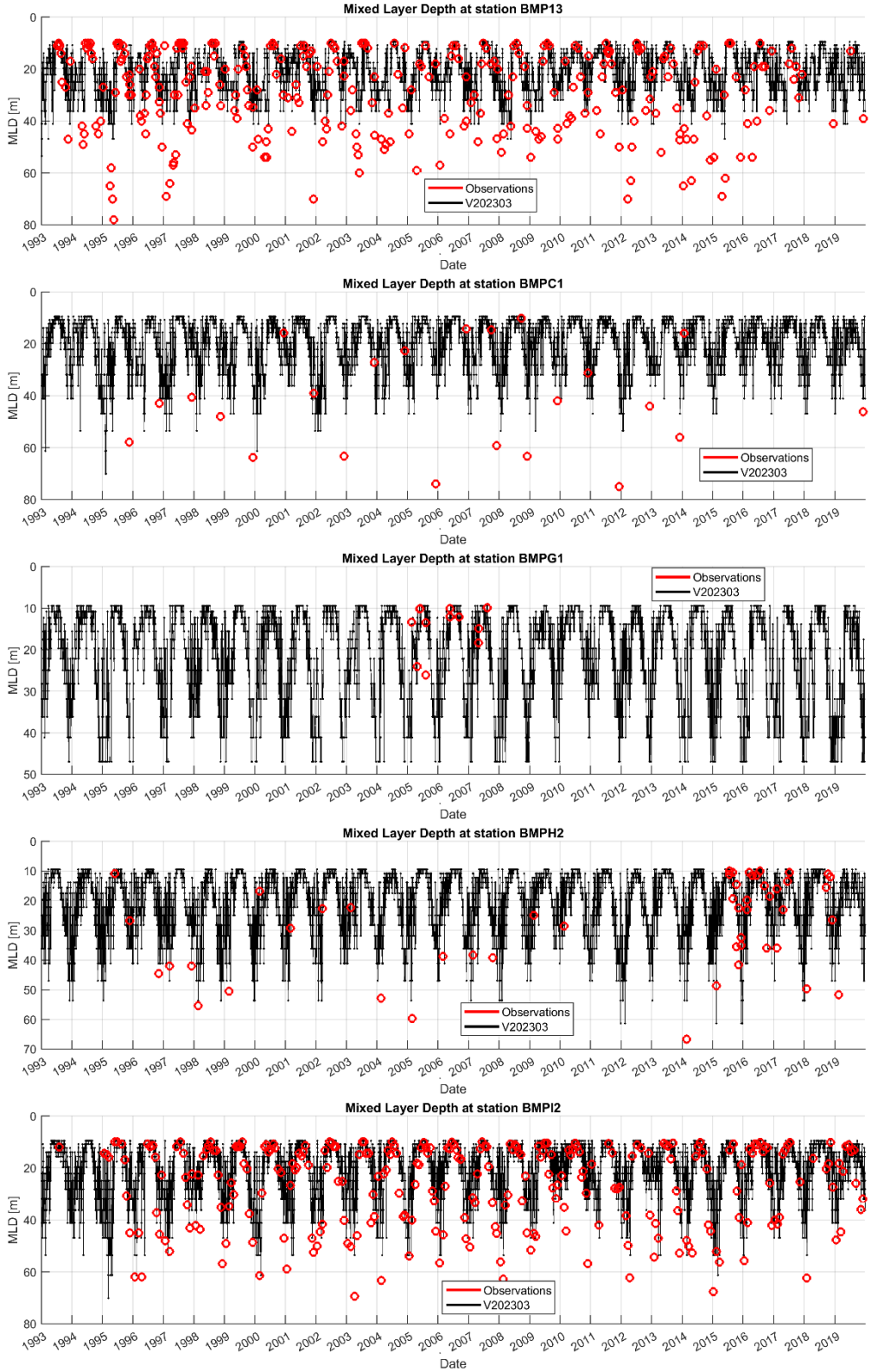


*Figure IV.5 Mixed layer depth time series at HELCOM stations in 1993-2019.*

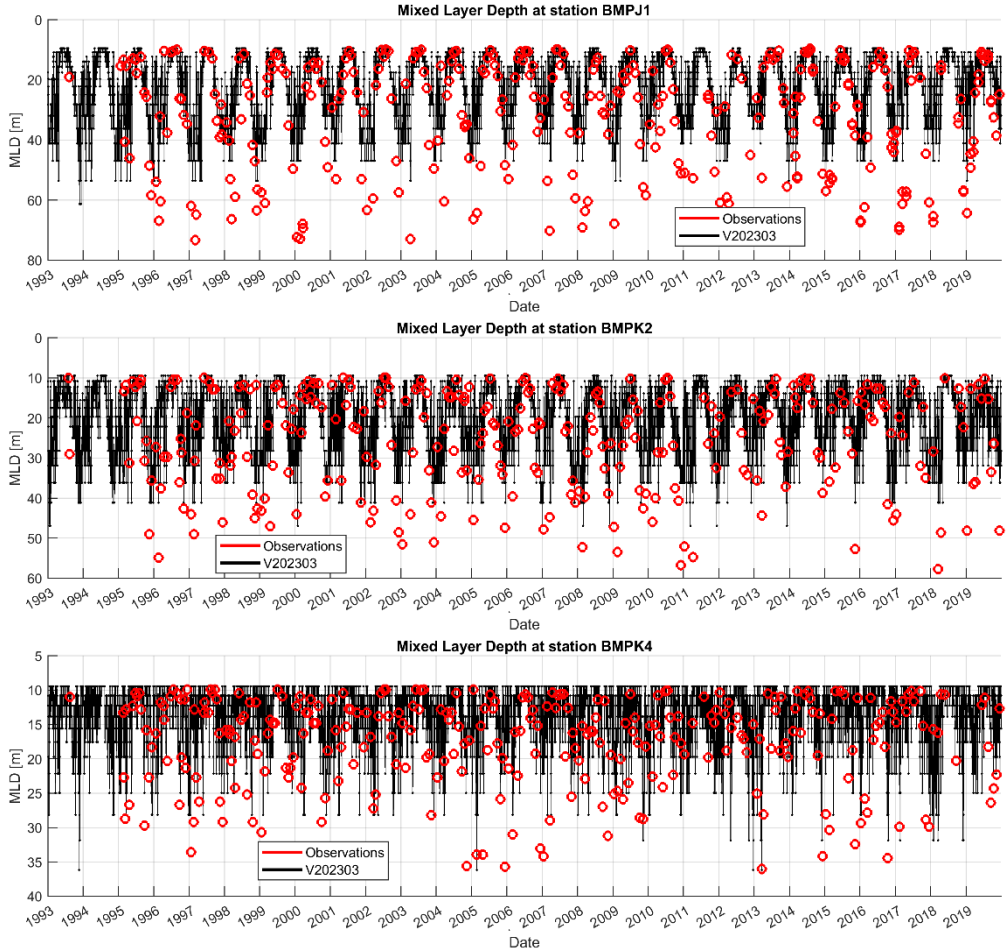


*Figure IV.6 Mixed layer depth time series at HELCOM stations in 1993-2019.*

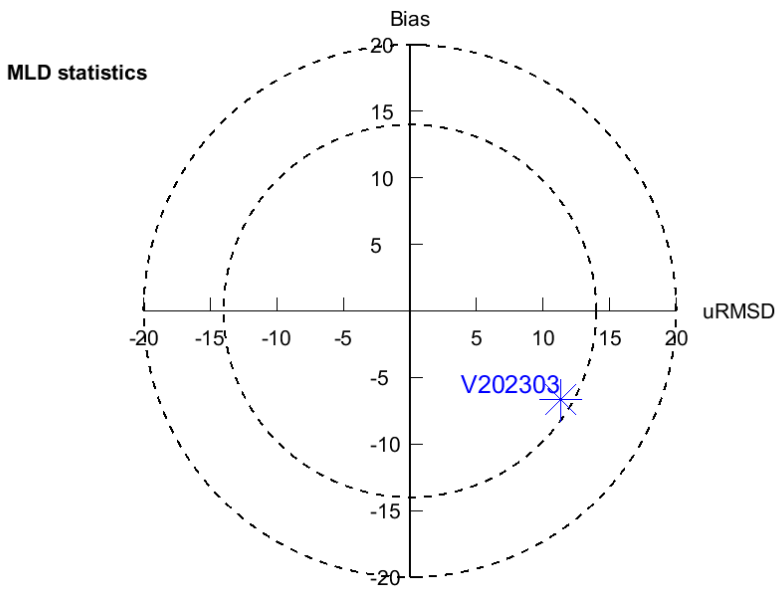


*Figure IV.7 Target diagram for the MLD at offshore grid points in 1993-2019, displayed in a Target diagram.*

## IV.2 Dissolved oxygen

Oxygen concentrations are validated with measurements at monitoring stations. For the geographical location of the stations, see Figure III.3. Figure IV.8 and Figure IV.9 summarise bias, cRMSD and correlation coefficient of oxygen concentrations at each station for the surface level and a deep level, respectively. Figure IV.10 gives an overview of the spatial and temporal distribution of oxygen concentrations at a selected monitoring station in the central Baltic Sea.

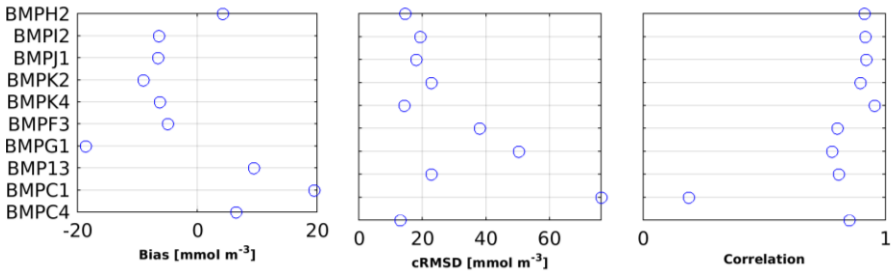


*Figure IV.8 Validation results of the MY product (blue circles) for the **surface level** at each oxygen monitoring station. Values of bias cRMSD and correlation coefficient represent time-averages of the period 01.01.1993 – 31.12.2018.*

The validation results at the monitoring stations indicate that oxygen concentrations at the surface are reproduced well by the model. At every station, the bias is low with values between – 20 and 20 mmol/m³ (Figure IV.8). The high correlation coefficients at the surface, as the well as the time series shown in Figure IV.10 (top panel) indicate that the seasonal patterns of oxygen concentrations are well reproduced by the model simulation.

In the deeper layers, the validation results show a strong positive bias at most stations (Figure IV.9), i.e. the oxygen concentrations in the deep levels are considerably overestimated by the model. In the deep basins of the Baltic Sea oxygen concentrations decline with depth to values close to zero. Such anoxic conditions in deep layers of the water column are typical for the central Baltic Sea, due to limited inflow of salty, oxygen rich water from the North Sea. This leads to a  a pronounced  stratification of the water column, where oxygen only reaches the deep waters through major inflow events from the North Sea. While the timing of the inflows of salty, oxygen rich water from the North Sea is well reproduced by the model (Figure IV.10), the decrease of oxygen concentrations after the inflow events is too slow. Therefore, at some stations the anoxic conditions are not well reproduced. This leads to the overall positive bias of in the deep basins of the Baltic Sea. If the anoxic conditions are not reproduced correctly by the model, it also affects the nitrogen and phosphorous dynamics in these regions (see sections IV.3 - IV.5). The overestimation of oxygen concentrations in the deep layers is connected to the shallow mixed layer depth of the physical model (see section IV.1), as well as an underestimation of organic matter concentrations in the sediments. We are continuously working to improve this issue in future versions.

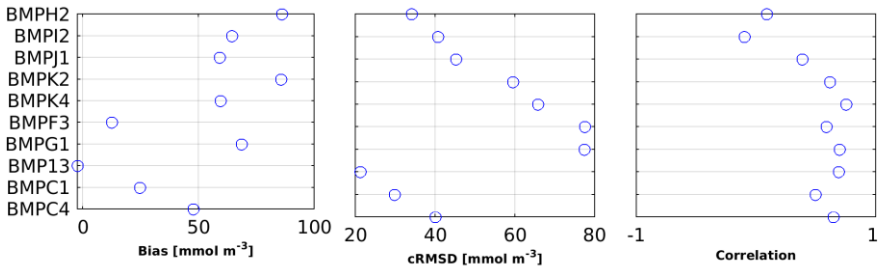


*Figure IV.9 Validation results of the MY product (blue circles) for the **deep level** at each oxygen monitoring station. Values of bias cRMSD and correlation coefficient represent time-averages of the period 01.01.1993 – 31.12.2018.*

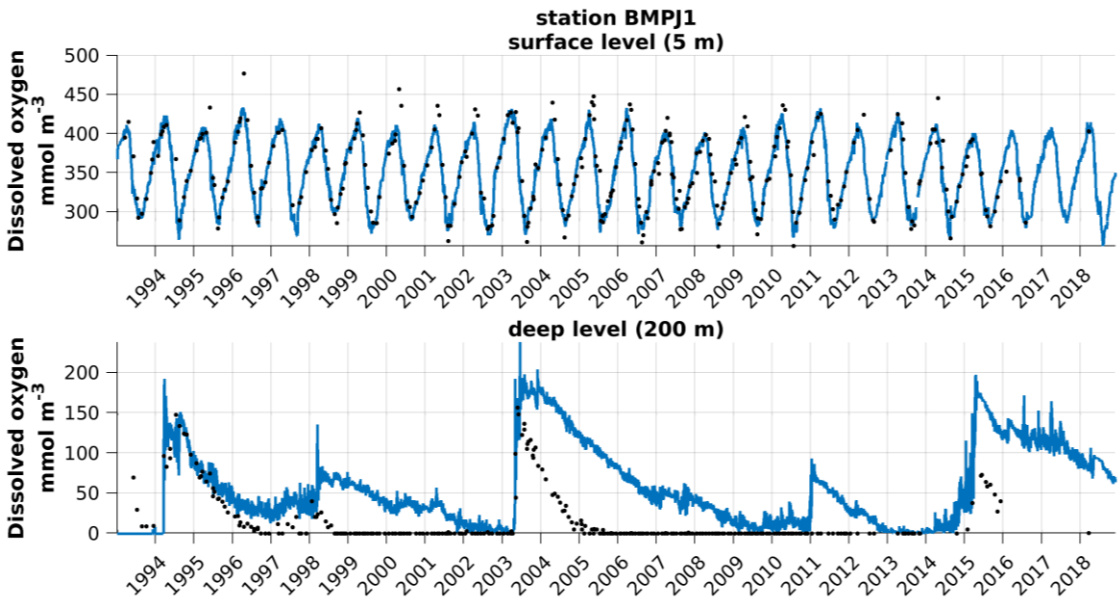


*Figure IV.10 Time series of oxygen concentrations at a station in the Baltic Proper (BMPJ1) for depth of 5 m and 200 m: black dots represent insitu measurements, blue lines the MY product. Time series and target diagrams for all oxygen monitoring stations are available on the [BOOS website](#).*

## IV.3 Ammonium

Ammonium concentrations are validated against measurements at monitoring stations. For the geographical location of the stations, see Figure III.3. Figure IV.11 and Figure IV.12 summarise bias, cRMSD and correlation coefficient of ammonium concentrations at each station for the surface level and the deepest level, respectively. Figure IV.13 gives an overview of the spatial and temporal distribution of ammonium concentrations at a selected monitoring station in the central Baltic Sea.

The validation results for ammonium show that concentrations are underestimated by the model throughout the whole water column. However, the deviation between the model simulation and in-situ measurements increases with depth. The negative bias is particularly strong in the deep basins of the Baltic Sea (Figure IV.12), where anoxic conditions are prevalent. Under anoxic conditions, ammonium accumulates, as it is not converted to nitrate. If the anoxic conditions are not reproduced by the

model, ammonium is converted to nitrate through nitrification, which leads to an underestimation of ammonium and an overestimation of nitrate concentrations in the anoxic regions. This connection is shown for the deep level at station BMPJ1, where measurements indicate anoxia: ammonium concentrations are underestimated (Figure IV.13), because oxygen concentrations are overestimated (Figure IV.10).

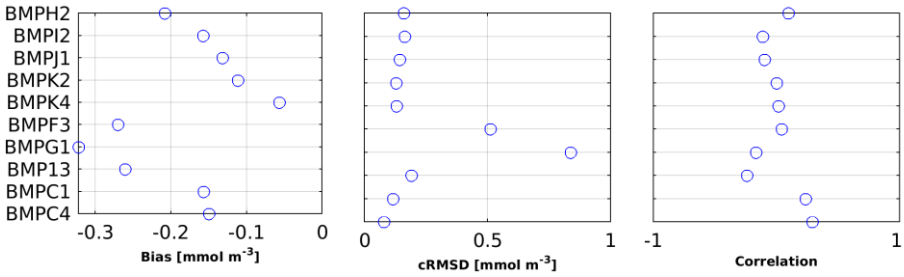


*Figure IV.11 Validation results of the MY product (blue circles) for the **surface level** at each ammonium monitoring station. Values of bias cRMSD and correlation coefficient represent time-averages of the period 01.01.1993 – 31.12.2018.*

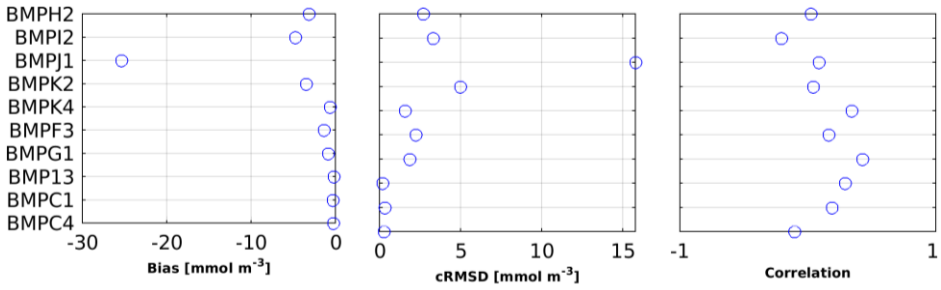


*Figure IV.12 Validation results of the MY product (blue circles) for the **deep level** at each ammonium monitoring station. Values of bias cRMSD and correlation coefficient represent time-averages of the period 01.01.1993 – 31.12.2018.*

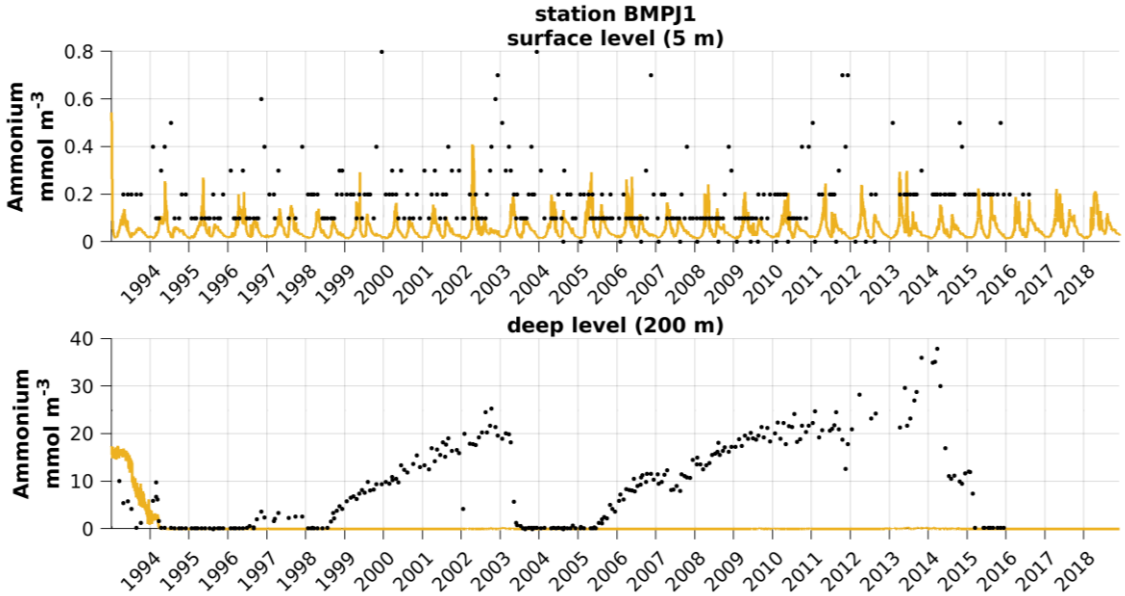


*Figure IV.13 Time series of ammonium concentrations at a station in the Baltic Proper (BMPJ1) for depth of 5 m and 200 m: black dots represent in-situ measurements, yellow lines the MY product. Time series and target diagrams for all ammonium monitoring stations are available on the [BOOS website](#).*

## IV.4 Nitrate

Nitrate concentrations are validated against measurements at monitoring stations. For the geographical location of the stations, see Figure III.3 Figure IV.14 and Figure IV.15 summarise bias, cRMSD and correlation coefficient of nitrate concentrations at each station for the surface level and the deepest level, respectively. Figure IV.16 gives an overview of the spatial and temporal distribution of nitrate concentrations at a selected monitoring station in the central Baltic Sea.

At the surface, validation results show a low bias between -1 and 0.5 mmol/m³ of nitrate concentrations at all monitoring stations (Figure IV.14). Further, Figure IV.16 (top panel) shows that the seasonal nitrate concentrations in the surface layer are simulated well by the model. In the deeper layers of the water column, nitrate concentrations are generally overestimated by the model (Figure IV.15 ). This is connected to oxygen concentrations in the deep basins. Nitrate is converted to nitrogen ($N_2$) under anoxic conditions (through denitrification). If oxygen concentrations are overestimated by the model and anoxic conditions are not simulated correctly, nitrate is also overestimated in these regions. This is particularly evident in the deep regions of Baltic Proper (e.g. stations BMPJ1, BMPK2 and BMPI2).

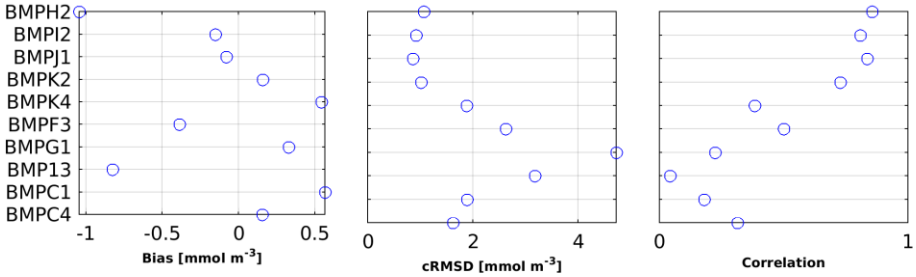


*Figure IV.14 Validation results of the MY product (blue circles) for the **surface level** at each nitrate monitoring station. Values of bias cRMSD and correlation coefficient represent time-averages of the period 01.01.1993 – 31.12.2018.*

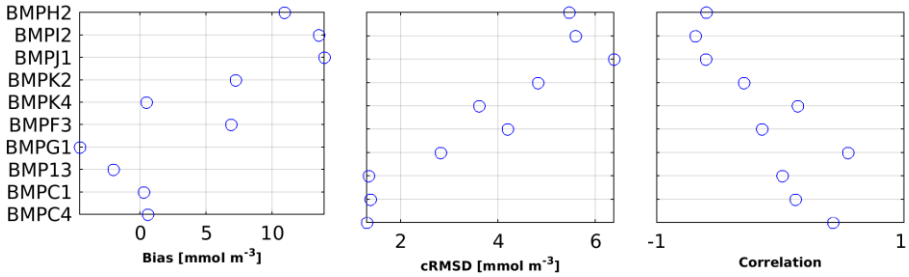


*Figure IV.15 Validation results of the MY product (blue circles) for the **deep level** at each nitrate monitoring station. Values of bias cRMSD and correlation coefficient represent time-averages of the period 01.01.1993 – 31.12.2018.*

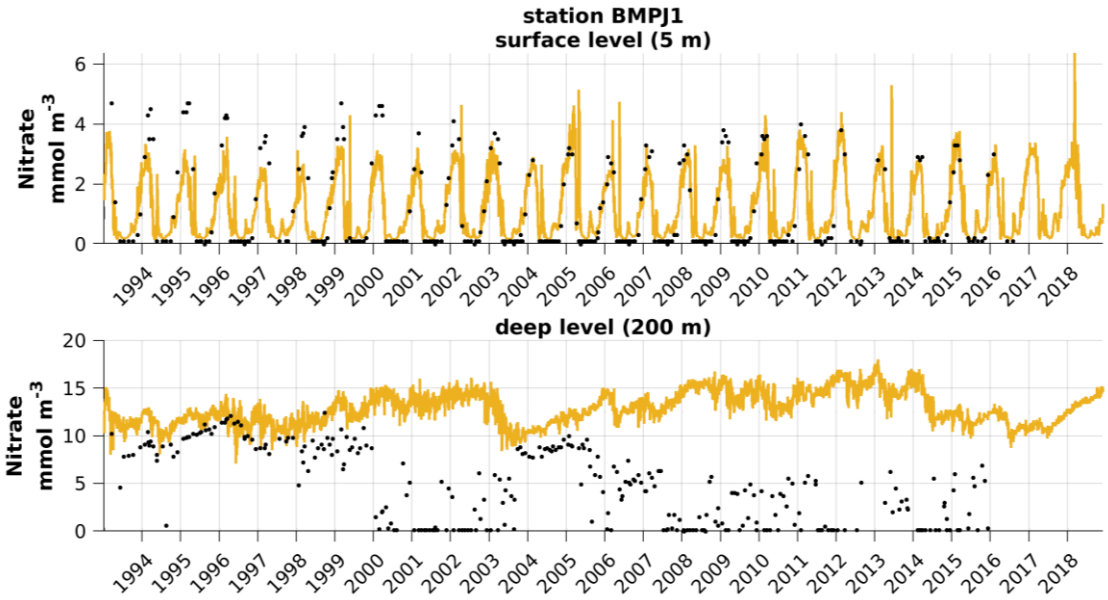


*Figure IV.16 Time series of nitrate concentrations at a station in the Baltic Proper (BMPJ1) for depth of 5 m and 200 m: black dots represent in-situ measurements, yellow lines the MY product. Time series and target diagrams for all nitrate monitoring stations are available on the [BOOS website](BOOS website).*

## IV.5 Phosphate

Phosphate concentrations are validated against measurements at monitoring stations. For the geographical location of the stations, see Figure III.3. Figure IV.17 and Figure IV.18 summarise bias, cRMSD and correlation coefficient of phosphate concentrations at each station for the surface level and the deepest level, respectively. Figure IV.19 gives an overview of the spatial and temporal distribution of phosphate concentrations at a selected monitoring station in the central Baltic Sea.

At the surface, the validation results for phosphate indicate a slight positive bias at the majority of the monitoring stations (Figure IV.17). At stations close to the coast, in the Gulf of Riga/Finland region (BMPG1, BMPF2) the bias is negative. As they are near to the coast, the phosphate concentrations at these stations are strongly influenced by nutrient input via rivers. Therefore, an underestimation of river phosphate loads might be the reason for the negative bias in the model results. Figure IV.19 shows that the seasonal dynamics of phosphate concentrations in the surface layer are reproduced well by the model simulation. In the deep layers of the water column, the phosphate concentrations are underestimated by the model at the majority of the monitoring stations (Figure IV.18). Figure IV.19 (bottom panel) shows that this negative bias is particularly strong in the deep basins of the Baltic Sea, where the anoxic conditions are common. The anoxic conditions lead to a release of phosphate from the sediments, which is adsorbed to iron-hydroxides. This process is not well represented by the model yet; therefore, phosphate concentrations in deep anoxic layers are underestimated.

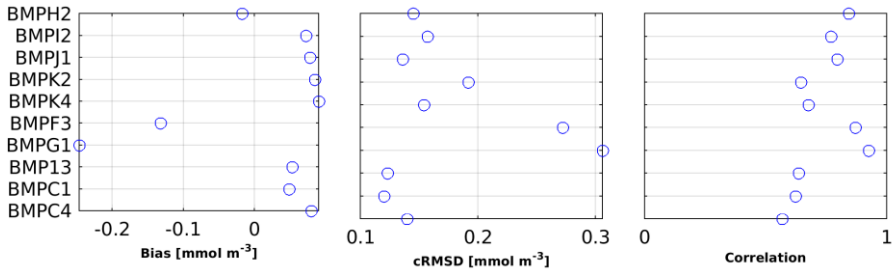


*Figure IV.17 Validation results of the MY product (blue circles) for the **surface level** at each phosphate monitoring station. Values of bias cRMSD and correlation coefficient represent time-averages of the period 01.01.1993 – 31.12.2018.*

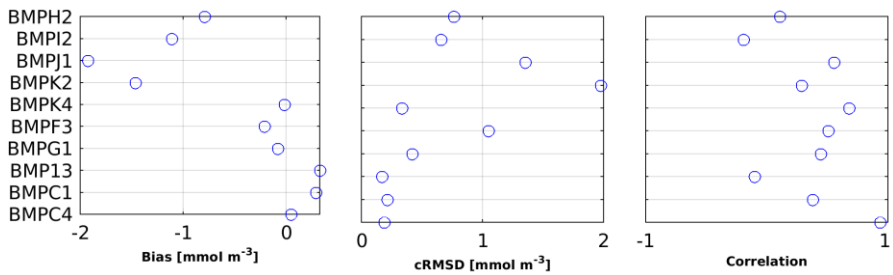


*Figure IV.18 Validation results of the MY product (blue circles) for the **deep level** at each phosphate monitoring station. Values of bias cRMSD and correlation coefficient represent time-averages of the period 01.01.1993 – 31.12.2018.*

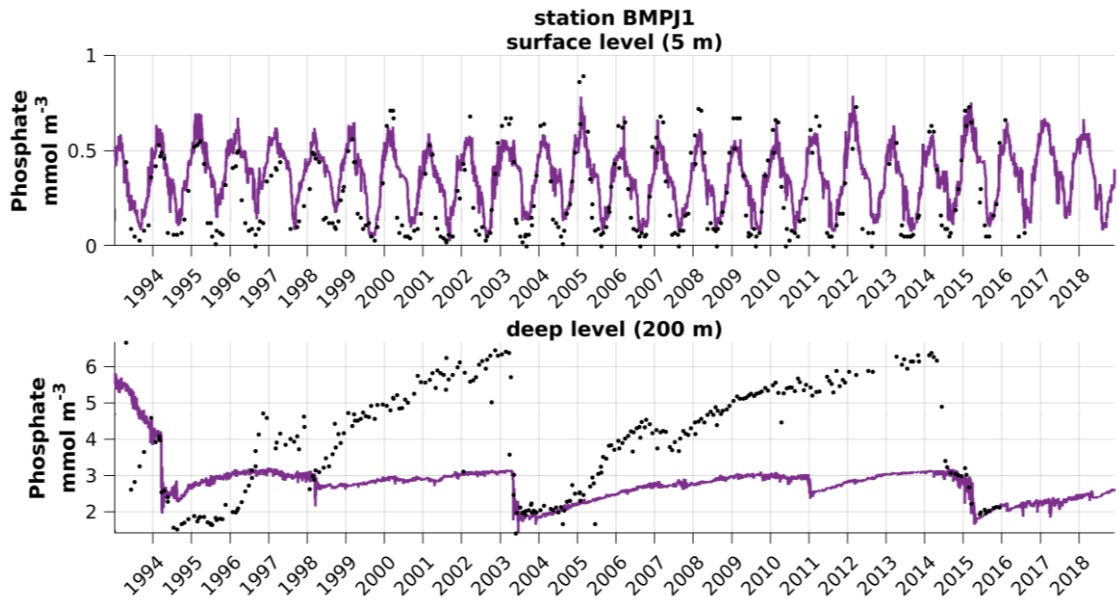


*Figure IV.19 Time series of phosphate concentrations at a station in the Baltic Proper (BMPJ1) for depth of 5 m and 200 m: black dots represent in-situ measurements, purple lines the MY product. Time series and target diagrams for all phosphate monitoring stations are available on the BOOS website.*

## IV.6 Chlorophyll a

Chlorophyll a concentrations are validated with measurements at monitoring stations. For the geographical location of the stations, see Figure III.3. Additionally, surface chlorophyll is validated with satellite observations. For that, monthly means of daily surface fields of instantaneous chlorophyll a concentrations were compared to satellite observations from product OCEANCOLOUR_BAL_BGC_L3_MY_009_133. The results for validation with the satellite observations is available on the BOOS website, it is not included in the statistical evaluation and calculation of the EANs presented here.

### Validation with station data

Figure IV.20 and Figure IV.21 summarise bias, cRMSD and correlation coefficient of chlorophyll a concentrations at each station for the surface level and the deepest level, respectively. Figure IV.22 gives an overview of the spatial and temporal distribution of chlorophyll a concentrations at a selected monitoring station in the central Baltic Sea.

The validation results show that chlorophyll a concentrations are underestimated at the surface and deeper layers in most regions of the Baltic Sea. Especially during the spring blooms of some years, the modelled chlorophyll a concentrations are lower than the measured values (Figure IV.22). This causes the overall negative bias shown in Figure IV.20 and Figure IV.21. The results shown in Figure IV.22 indicate that the seasonal dynamics and interannual variability of chlorophyll a concentrations are reproduced by the model. The comparison with the in-situ measurements shows that the timing of the spring phytoplankton bloom is simulated well by the model. In some years (e.g. 2015), the model produces chlorophyll peaks in summer, which cannot be observed in the in-situ observations. These pronounced summer blooms seem to be caused by cyanobacteria blooms imprecisely simulated by the model.

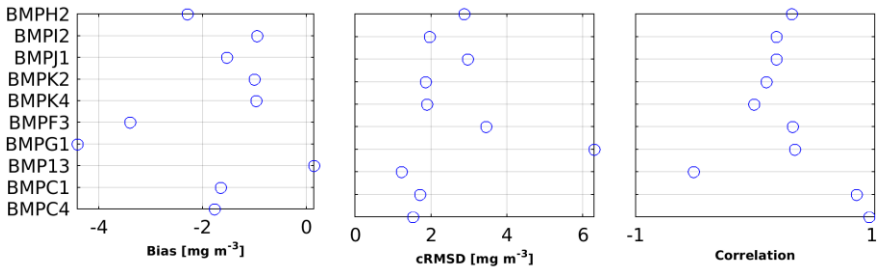


*Figure IV.20 Validation results of the MY product (blue circles) for the **surface level** at each chlorophyll monitoring station. Values of bias cRMSD and correlation coefficient represent time-averages of the period 01.01.1993 – 31.12.2018.*

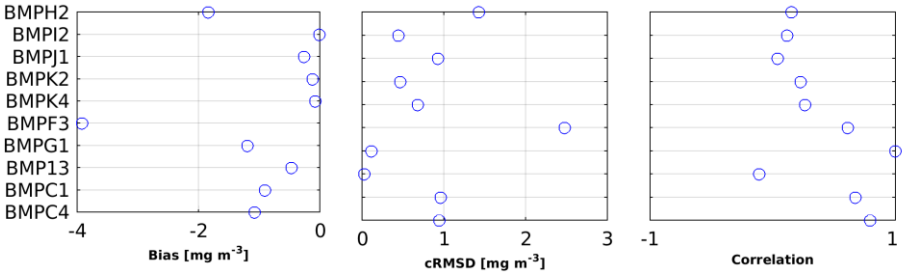


*Figure IV.21 Validation results of the MY product (blue circles) for the **deep level** at each chlorophyll monitoring station. Values of bias cRMSD and correlation coefficient represent time-averages of the period 01.01.1993 – 31.12.2018.*

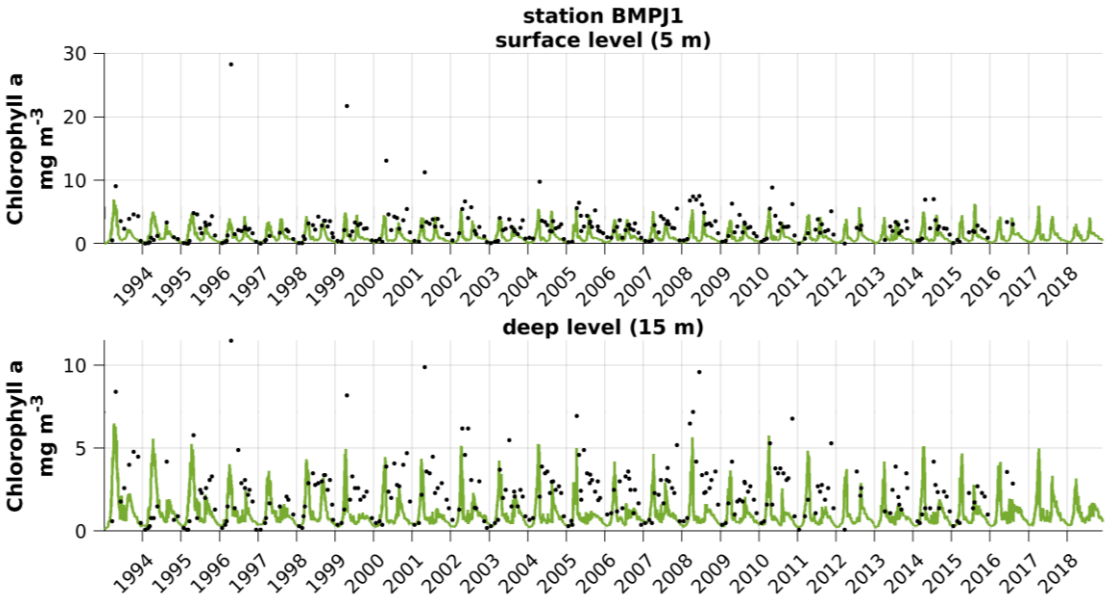


*Figure IV.22 Time series of chlorophyll a concentrations at a station in the Baltic Proper (BMPJ1) for depth of 1 m and 50 m: black dots represent in-situ measurements, green lines the MY product. Time series and target diagrams for all chlorophyll monitoring stations are available on the [BOOS website](BOOS website).*

## IV.7 Net primary production

For net primary production, no observational data is available for the validation period in the Baltic Sea. However, to give an overview of the temporal and spatial dynamics of net primary production simulated by the model. Results for a selected station in the central Baltic Sea are shown in Figure IV.23.

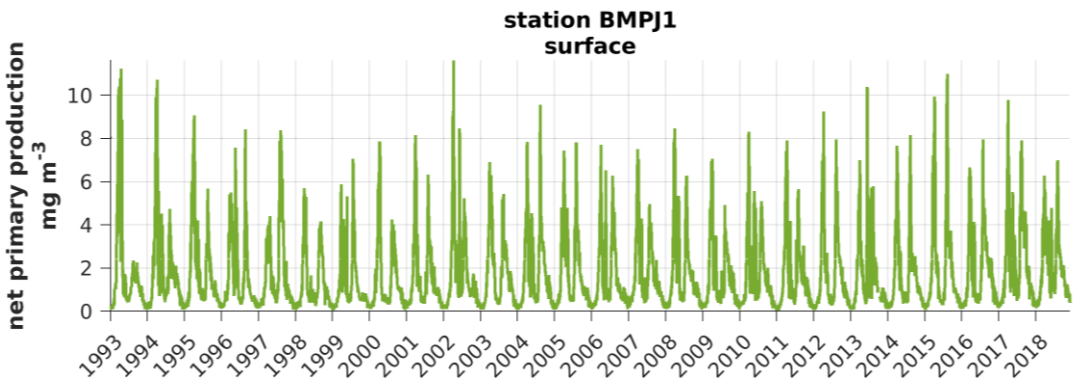


*Figure IV.23 Time series of surface net primary production at a station in the Baltic Proper (BMPJ1): green lines represent the MY product. Time series for all monitoring stations are available on the [BOOS website](#).*

## IV.8 Secchi depth

Secchi depth is validated against measurements at monitoring stations. For the geographical location of the stations, see Figure III.3. Figure IV.24 summarises bias, cRMSD and correlation coefficient of the Secchi depth at each station. Figure IV.25 gives an overview of the seasonal dynamics of the Secchi depth at a selected monitoring station in the Baltic Proper (BMPJ1).

The validation results show that simulated values of Secchi depth match the measured values reasonably well at all monitoring stations. However, the positive bias at all monitoring stations (Figure IV.24) indicates that the model overestimated Secchi depth in all regions of the Baltic Sea. This is consistent with the results for chlorophyll concentrations. The model underestimates chlorophyll, i.e. a lower turbidity is assumed in the model simulations, which results in a higher Secchi depth. Figure IV.25 shows that the general seasonal dynamics of Secchi depth are reproduced by the model. An exception are very high observed values for Secchi depth during winter, which are not captured in the simulation.

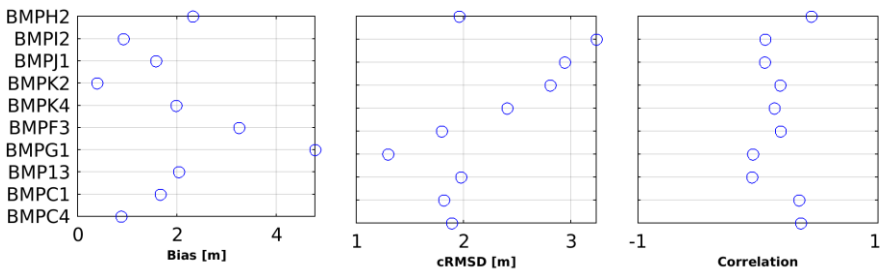


*Figure IV.24 Validation results of the MY product (blue circles) at each Secchi depth monitoring station. Values of bias cRMSD and correlation coefficient represent time-averages of the period 01.01.1993 – 31.12.2018.*

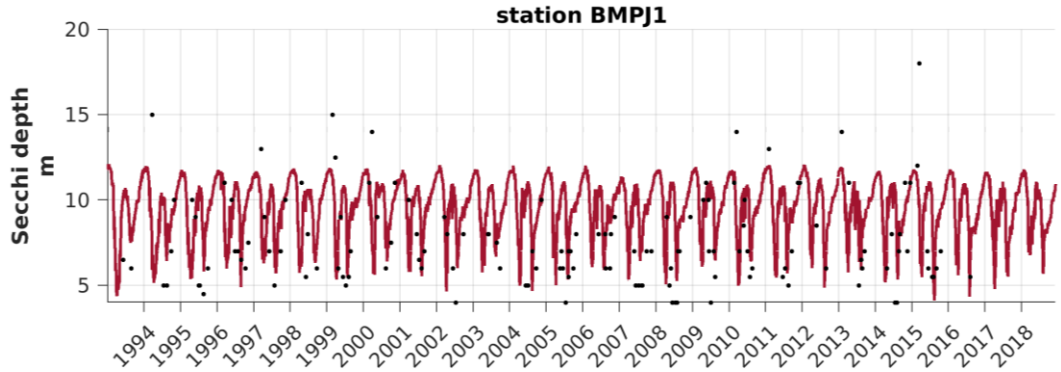


*Figure IV.25 Time series of Secchi depth at a station in the Baltic Proper (BMPJ1): black dots represent in-situ measurements, red lines the MY product. Time series and target diagrams for all Secchi depth monitoring stations are available on the BOOS website.*

## IV.9 pCO2

The partial pressure of carbon dioxide (pCO2) at the surface is validated with measurements from the in-situ ferrybox observations obtained from the SOCAT database. For the reanalysis period Jan 2012 to Dec 2019, data from the Finnmaid ferrybox is available (*Figure* III.4). The forecasts were extracted from the product mid-night surface layers by linear interpolation to the observation locations and times and the statistics were calculated using the obtained time series.

Figure IV.26 presents the ferrybox observations and simulation results for the extractions from the product V202303 for the reanalysis period 2012-2019. The system V202303 shows higher partial pressure of carbon dioxide each year in spring and in the beginning of the summer. The reason for this bias might be the atmospheric forcing used, which is a global climatology.

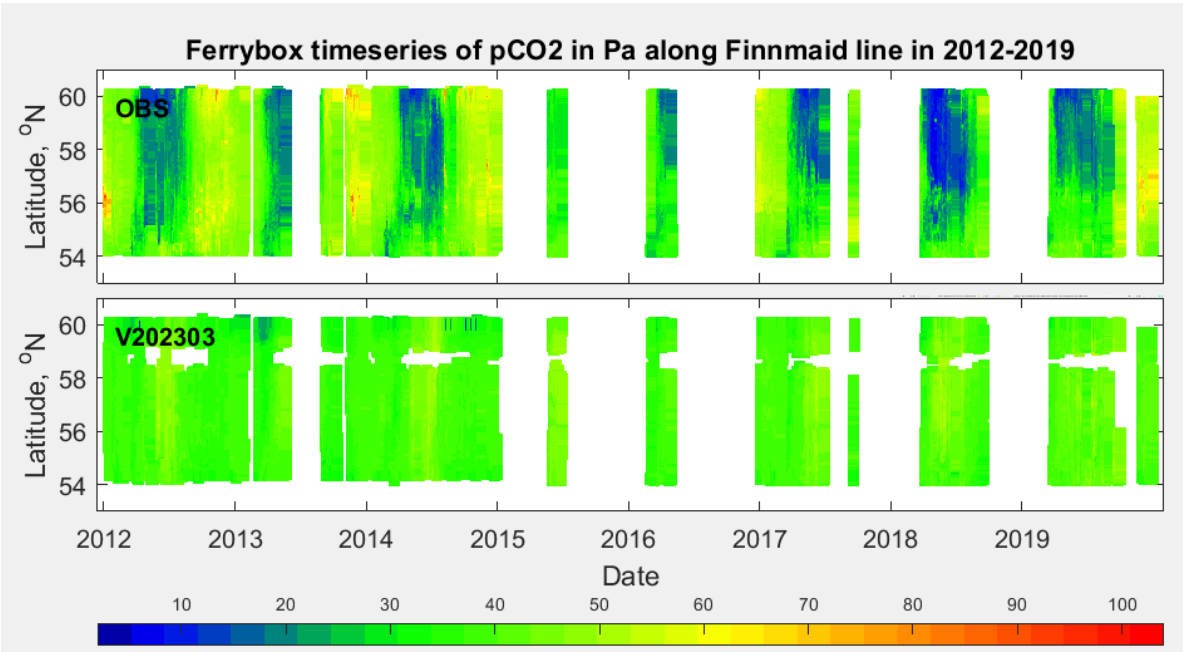


*Figure IV.26 Observed (OBS) and product (V202303) partial pressure of carbon dioxide in Pascal are shown for transects of the Finnmaid line for the period 2012-2019.*

The product deviation is presented similarly with observations with time-latitude maps in Figure IV.27. The deviation plots reveal that product V202003 overestimates the partial pressure of carbon dioxide each year in spring and in the beginning of the summer.

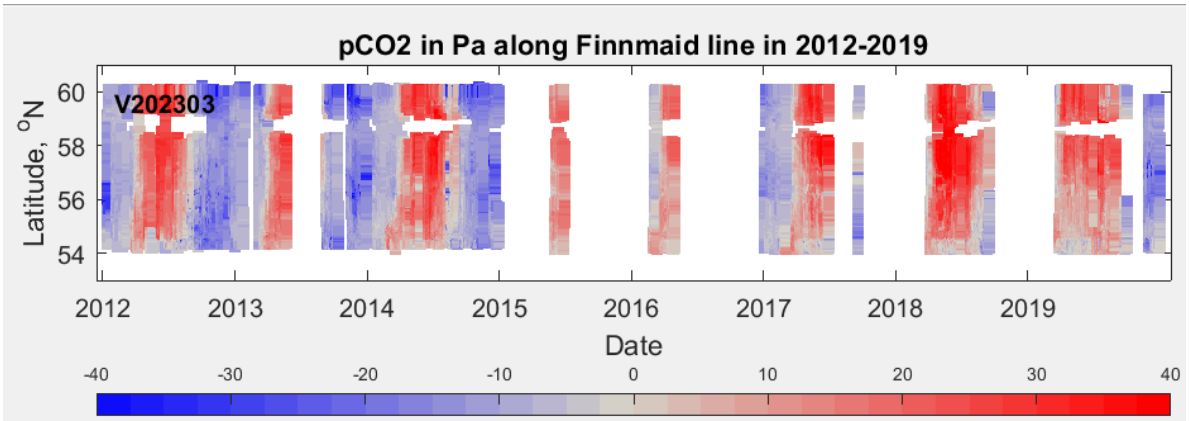


*Figure IV.27 Partial pressure of carbon dioxide deviation along the Finnmaid line for the best estimate (+12 hours forecast) for the period 2012-2019.*

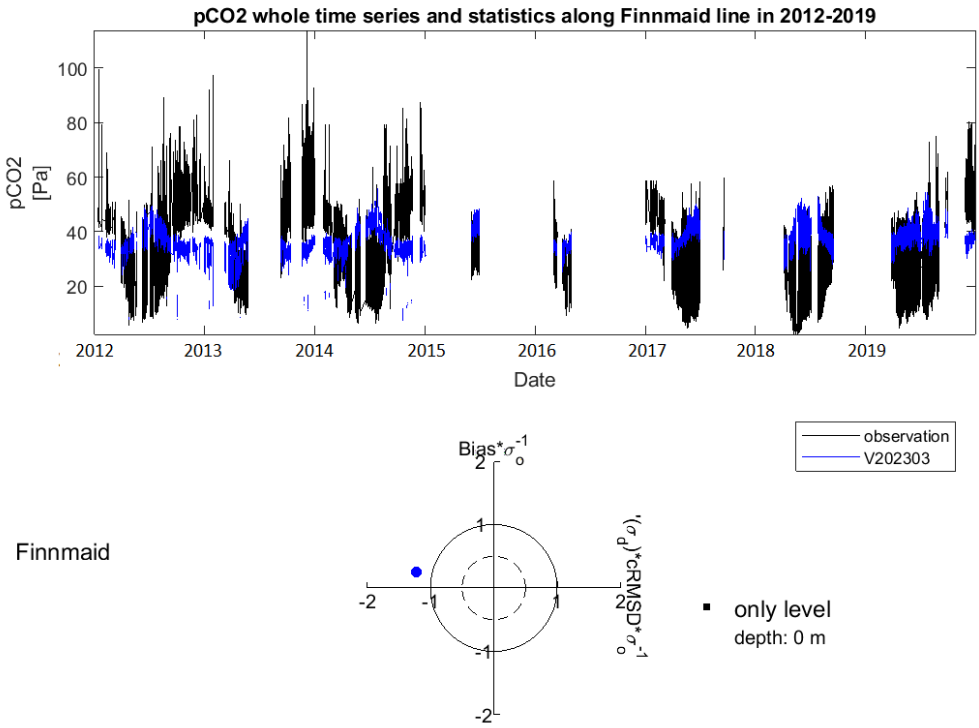


*Figure IV.28 Partial pressure of carbon dioxide statistics are plotted for transects of the Finnmaid line. At the top, the 2012-2019 time series is shown. At the bottom, the Target diagram is shown for the best estimate.*

Figure IV.28 presents the time series of measurements and V202303, as well as the statistics for the period, in form of a target diagram. Product V202303 does not show a good agreement with the observations in terms of general statistics.

## IV.10 pH

PH is validated against measurements at monitoring stations. Figure IV.29 and Figure IV.30 summarise bias, cRMSD and correlation coefficient of pH values at each station for the surface level and the deepest level, respectively. Figure IV.31 gives an overview of the spatial and temporal distribution of oxygen concentrations at a selected monitoring station in the central Baltic Sea.

At the surface, the validation results show a negative bias for pH at all Baltic Sea Regions, except for the Bothnian Bay, where the bias is positive. For the deeper level, the bias increases and there is a positive bias for pH values at the majority of monitoring stations. Figure IV.31 shows that the model simulates seasonal patterns of pH at the surface. However, the variability is much lower than for the in-situ observations. In very deep layers, pH values are consistently underestimated by the model over the entire reanalysis period. Overall, the model is able to simulate the general temporal and spatial patterns of pH, whereas the pH at the surface are matched better by the model than in deep parts of the Baltic Sea.

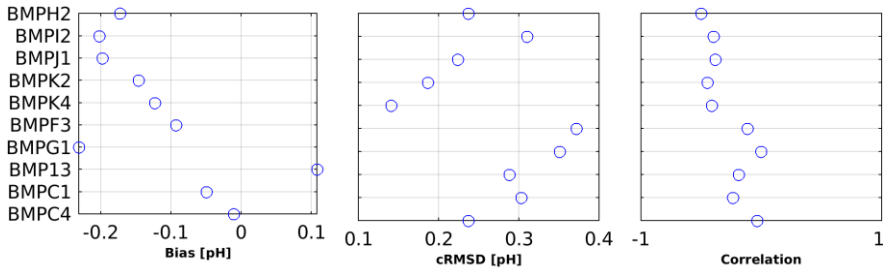


*Figure IV.29 Validation results of the MY product (blue circles) for the **surface level** at each pH monitoring station. Values of bias cRMSD and correlation coefficient represent time-averages of the period 01.01.1993 – 31.12.2018.*

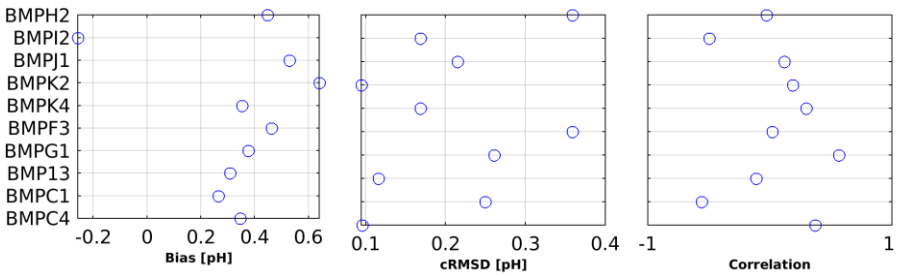


*Figure IV.30 Validation results of the MY product (blue circles) for the **deep level** at each pH monitoring station. Values of bias cRMSD and correlation coefficient represent time-averages of the period 01.01.1993 – 31.12.2018.*

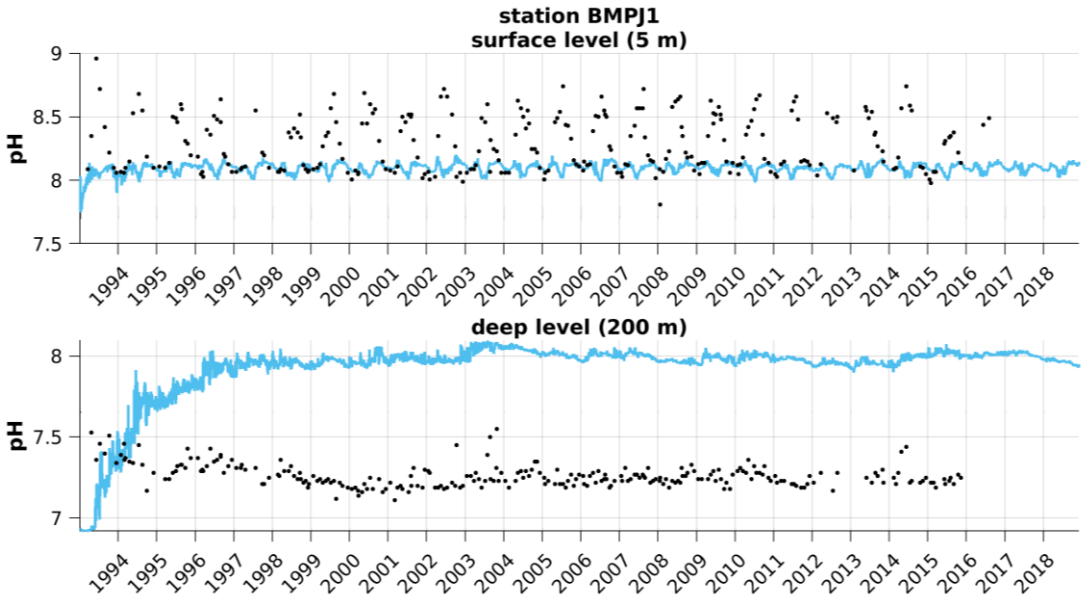


*Figure IV.31 Time series of pH concentrations at a station in the Baltic Proper (BMPJ1) for depth of 1 m and 200 m: black dots represent in-situ measurements, blue lines the MY product. Time series and target diagrams for all pH monitoring stations are available on the [BOOS website](#).*

# V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

## V.1 Events and changes

| Date | Change/Event description | System version | other |
|---|---|---|---|
| 2019 | New reanalysis was released covering years 1993 to 2018. The reanalysis was produced with Nemo3.6-Scobi model system and assimilation of physical and biogeochemical profiles. Horizontal resolution of the product is 2 nm. | 1.0 | |
| 2023 Extended in 2024 | New reanalysis covering years 1993 to 2024 produced with new model system: Nemo4.0-ERGOM and assimilation of SST, S/T profiles with PDAF. Horizontal resolution 1 nm. | 2.0 | |

## V.2 $NO_3$ values in river run-off data from the EHYPE model

In the course of the year, unrealistically high nitrate concentrations occur at a few flat grid points near the coast. This is directly related to the E-hype data sets used. At very low fresh water discharge rates of certain E-hype outlet points, the riverside nitrate concentrations increase strongly, which leads to artificially unrealistically high nitrate concentrations at particularly flat grid points in the model. This effect is accepted for the time being, since the high nitrate concentrations are limited to single grid points due to the low amounts of substances and do not spread further in the model area.

## V.3 $pCO_2$ values near river mouth

Near some river mouth, very high $pCO_2$ concentrations of up to 1000 Pa are sometimes reflected in the product. This happens especially at low salinity levels near very eutrophic estuaries. Even if these concentrations in the product cannot be confirmed in the absence of suitable observation data, these concentration maxima are by no means unrealistic. Nevertheless, it should be noted that the calculation method of the carbonate system in the model used is not parameterised to estuaries.

# VI QUALITY CHANGES SINCE PREVIOUS VERSION

The new product is based on a completely new model system. The Nemo3.6-Scobi system was replaced by the Nemo4.0-ERGOM system, which is also used for the BAL MFC near-real-time product. The horizontal resolution is increased from 2 to 1 nautical miles.

With the new model system, we were able to improve the following issues:

- Very high positive biases of nitrate and phosphate concentrations in the Gulf of Riga and Gulf of Bothnia regions are significantly improved. This was achieved by calibrating the river loads of nitrate and phosphate entering the Baltic Sea in these regions.
- More realistic benthic-pelagic coupling, by parameter optimisation of the sediment burial process and detritus sinking velocity.
- A carbon cycle was introduced with the new model system; therefore, pH and pCO2 are available as new products.

Note that in the previous Nemo-Scobi system, physical and biogeochemical parameters were assimilated, while in the new Nemo-ERGOM only SST and T/S are assimilated. That might cause a slight deterioration of quality in the biogeochemical products in some regions.

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